

MedAid — Safety-First Multimodal AI Triage for Rural Healthcare

Rapid, reliable first-line guidance using rules + LLM JSON

Team Members

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Problem & Opportunity

- Rural/semi-urban users struggle to interpret symptoms/reports
- Low health literacy → delays, missed emergencies
- Travel costs vs. risk of waiting
- Need: safe first step that is always available and understandable

Personas

- Farmer with night-time chest discomfort, 40km from clinic
- Elderly patient with confusing lab report
- Young mother with toddler fever & vomiting

Key Objectives

- Detect abnormal values from reports
- Store history for personalization
- Understand symptoms (text/audio)
- Predict likely conditions
- Provide risk-based actions (ER vs. clinic vs. self-care)

Dataset Link :-

Dataset Details : Nhanes dataset , DDXPlus dataset

Introduction

Problem :

- Rural populations face limited access to timely healthcare due to sparse providers, poor transport, and weak infrastructure.
- Early triage is critical to differentiate benign from life-threatening conditions but current tools are inadequate.
- Limitations of Existing Systems
- Rule-based symptom checkers → rigid decision trees, high over-triage rates.
- Pure LLM-based chatbots → flexible reasoning but unsafe/misleading outputs.

Proposed Solution - MedAid

Key Objectives Detect abnormal values from reports Store history for personalization Understand symptoms (text/audio) Predict likely conditions Provide risk-based actions (ER vs. clinic vs. self-care)

System Architecture

1. **User Interface** – Streamlit web app captures text or medical report images.
2. **Input Processing** – Text normalization, OCR for lab data.
3. **Rule-Based Safety Checker** – Deterministic emergency triggers (e.g., SpO₂ < 90%, “chest pain radiating to arm”) immediately escalate to Emergency.
4. **Prompt Builder & LLM (Gemini)** – Generates structured JSON output with conditions, risk levels, probabilities, reasoning, and recommendations.
5. **Geolocation Module** – Uses Google Places / OpenStreetMap to suggest nearby facilities.
6. **Database Layer** – MongoDB/JSON stores patient history, triage outputs, and consultation records for continuity of care.

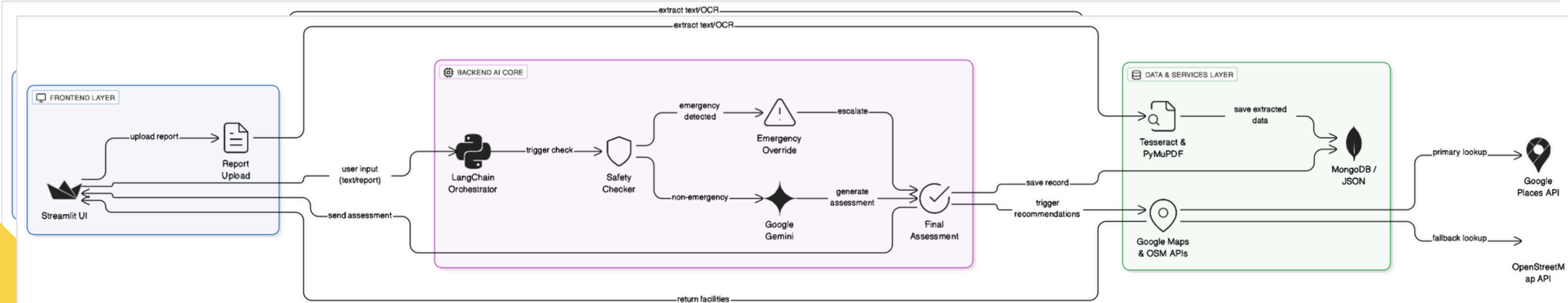


Figure 2: Architecture Diagram

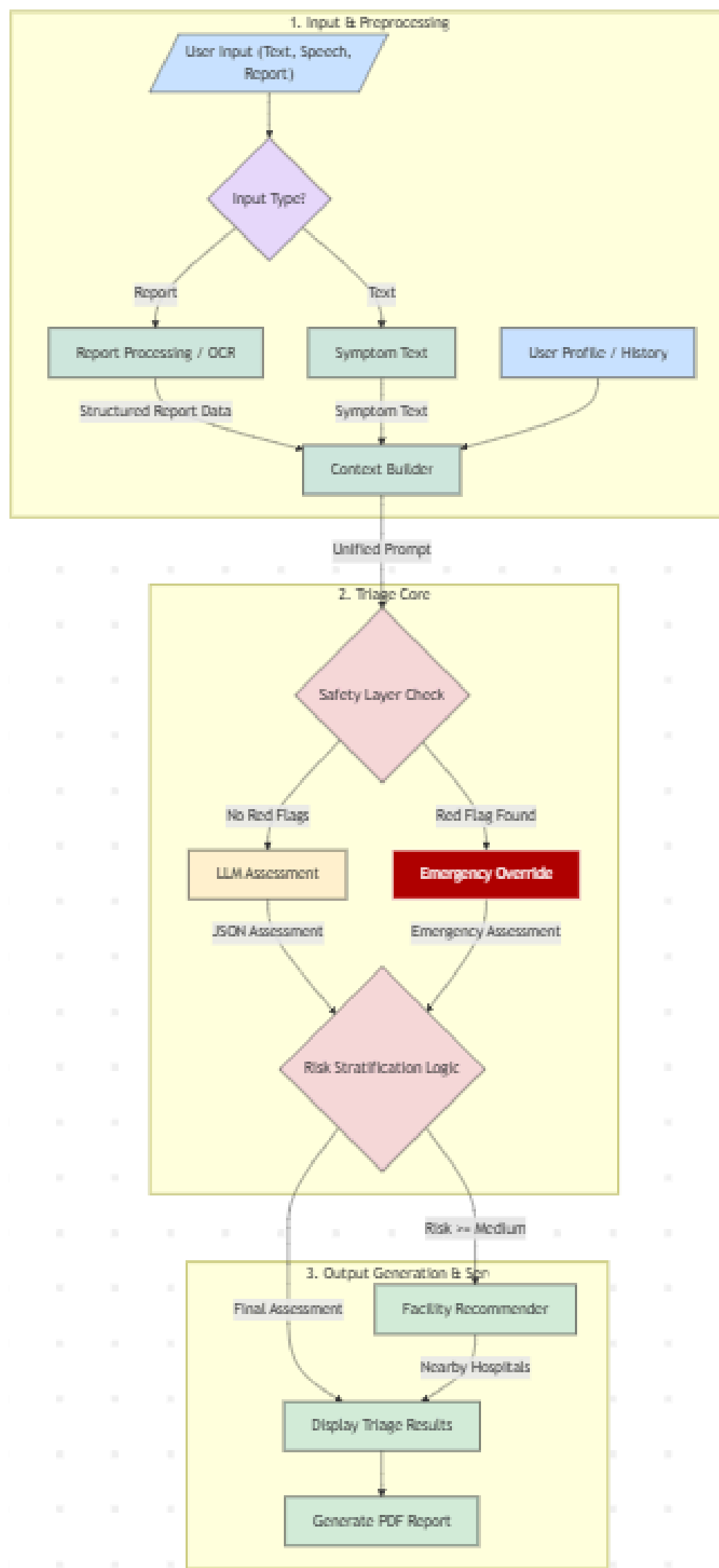


Figure 5: FlowChart

Literature Survey

Survey on Methodologies

- Roustan, D., Lee, J. J., & Topol, A. (2025). The Clinicians' Guide to Large Language Models. npj Digital Medicine, 8(1).
- Singhal, K., et al. (2025). Toward expert-level medical question answering with large language models. Nature Medicine.
- Singhal, K., et al. (2023). Large language models encode clinical knowledge. Nature.

Survey on Datasets

- Heidelberg Institute for Geoinformation Technology (HeiGIT). (2021–2022). Insights into OpenStreetMap healthcare attributes in India.
- G. Jain et al. (2025). Geo-mapping of government healthcare facilities providing diagnostic services in Delhi. BMJ Public Health.
- OSM_RMSI. (2020). Our Experience working with Health Facilities Import—India. OpenStreetMap Diary.

Survey on Gaps

- E. Asgari et al. (2025). A framework to assess clinical safety and hallucination rates of LLMs for medical text summarisation. NPJ Digit. Med.
- D. McDuff et al. (2025). Towards accurate differential diagnosis with large language models.
- Y. Kim et al. (2025). Medical Hallucination in Foundation Models and Their Implications.

Survey on results

- Masanneck, L., Balakrishnan, R., & Desborough, J. (2024). Triage Performance Across Large Language Models and Clinicians. JMIR.
- C. Y. K. Williams et al. (2024). Use of a Large Language Model to Assess Clinical Acuity of Adults in the Emergency Department. JAMA Netw. Open.

Details

Tech Stack

- Large Language Model: Google Gemini
- OCR Engine: Tesseract + PyMuPDF
- Backend & Frontend: Python (Streamlit) for UI and processing pipeline.
- Database: MongoDB + local JSON for patient history
- Geo-Mapping: Google Places API + OpenStreetMap

Innovativeness

- **Safety-First Hybrid Architecture:** The project's most novel contribution is its hybrid design that combines a deterministic, rule-based safety layer for emergency detection with a flexible, generative LLM (Gemini) for nuanced reasoning.
- **Multimodal Input Handling:** The system is designed to accept diverse inputs, including typed text, and structured data extracted from medical reports via OCR, making it more accessible and comprehensive.
- **Hyper-Local Recommendations:** The system provides contextually relevant guidance by using open-source map data (OpenStreetMap) to find and suggest the closest healthcare facilities, which is a critical feature for rural deployment.

Output/Results

- **Feasibility and Trustworthiness:** The project demonstrates that a hybrid architecture makes AI triage both feasible and trustworthy for rural healthcare environments by balancing the flexibility of an LLM with the reliability of deterministic rules.
- **Improved Safety and Accuracy:** The pilot evaluation shows that Medaid is superior to both rule-only and LLM-only systems, as it successfully identifies all emergencies (1.00 recall) while significantly reducing the rate of over-triage.
 - **Enhanced Healthcare Access:** The system has the potential to improve healthcare delivery in underserved areas by enabling earlier recognition of emergencies and providing tailored guidance for less urgent conditions, thus helping to better allocate scarce medical resources.